

Multimodal Depression Detection

in Dravidian Languages (Tamil & Malayalam)

Course: **22AIE315 Natural Language Processing**

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github.com/S6-Project-Workspace/depression-detection-from-speech

Problem Statement & Motivation

WHO: Over 280 million people globally suffer from depression

Code-mixing complexity

Speakers mix Tamil/Malayalam with English, breaking standard NLP models.

Limited labeled datasets

Scarce annotated mental health data for low-resource Dravidian languages.

Linguistic complexity

Highly agglutinative morphology — each word encodes tense, person, and voice.

KEY CHALLENGES

Code-Mixing

Tamil + English mixed text

Dialect Variation

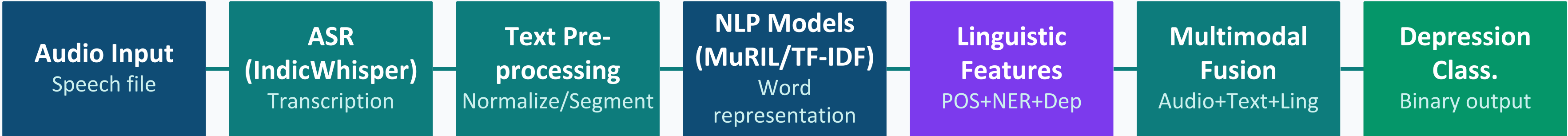
Regional linguistic gaps

Limited Data

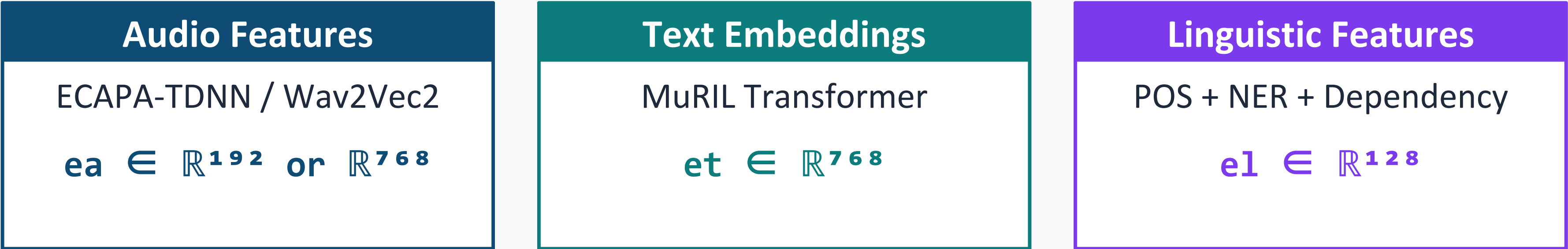
Low-resource barrier

Linguistic complexity

Complex word formation



INPUT MODALITIES



$$e_{\text{fused}} = [e_a; e_t; e_l] \in \mathbb{R}^{192 + 768 + 128} = \mathbb{R}^{1088}$$

Clean

Remove special chars,
extra whitespace, noise

Normalize

Unicode NFC +
Indic normalization

Language ID

Tamil / Malayalam /
Code-mixed detection

Segment

Morfessor MDL-based
morphological
segmentation

Unicode Normalization (NFC) - Normalization Form Canonical Composition

$$\text{NFC}(s) = \text{compose}(\text{decompose}(s))$$

decompose(s) splits characters into base + combining marks.
compose recombines into canonical Unicode form.

Morfessor MDL Segmentation

splits words into morphemes → chooses the best split

$$\text{Cost}(M, D) = L(M) + L(D | M)$$

$$\text{seg}^*(w) = \arg \min_{\text{seg}} \sum_{m \in \text{seg}} -\log P(m)$$

வருகிறேன் → வர் + கிற + ஏன் = [come]+[pres]+[1st person]

TF-IDF (Baseline) — Macro-F1: 0.72

$$TF(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}$$
$$IDF(t, D) = \log \frac{|D|}{|\{d \in D : t \in d\}|}$$

$$TF-IDF(t, d, D) = TF(t, d) \times IDF(t, D)$$

SVM optimization objective:

$$\min_{w, b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i$$

$$\text{s.t. } y_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

Limitation: Cannot capture contextual semantics.
"I feel pain" vs "pain feels like silence" — identical bag-of-words.

Measures the importance of a word in a document

MuRIL Transformer — Macro-F1: 0.81

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

$$\text{FFN}(x) = \text{GELU}(xW_1 + b_1) W_2 + b_2$$

12 layers | 768 hidden dims | 17 Indian languages

$$\text{logits} = \text{Dropout}(h_{CLS}) W_c + b_c$$

$$\mathcal{L}_{CE} = - \sum_{i=1}^N \sum_{c=1}^C y_{ic} \log(\hat{y}_{ic})$$

Macro-F1 improves from 0.72 → 0.81. Self-attention captures long-range linguistic dependencies.

Traditional NLP Tasks

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Linguistic feature extraction — POS, NER, Dependency Parsing

POS Tagging

17 UD tags — CRF sequence labeling

$$P(y \mid x) = \frac{1}{Z(x)} \exp\left(\sum_i \sum_k \lambda_k f_k(y_{i-1}, y_i, x, i)\right)$$

$$y^* = \arg \max_y P(y \mid x) \quad [\text{Viterbi decoding}]$$

$$H(P) = - \sum_{t \in T} P(t) \log_2 P(t) \quad [\text{POS diversity}]$$

Signal: elevated first-person pronoun ratio correlates with self-focus in depression.

Named Entity Recognition

BIO tagging — PERSON, MEDICAL, LOC

$$P(y \mid x) = \frac{1}{Z(x)} \prod_{t=1}^n \psi_t(y_{t-1}, y_t, x)$$

$$R_{med} = \frac{|\text{Medical Entities}|}{|\text{All Entities}|}$$

$$R_{1st} = \frac{|\{w : w \in \text{pronouns}\}|}{|W|}$$

Signal: high medical entity ratio and first-person entity count indicate health concerns.

Dependency Parsing

Universal Dependencies — SOV structure

$$T^* = \arg \max_T \sum_{(i,j,r) \in T} s(i,j,r)$$

$$C_{syn} = 0.4 \cdot \text{depth}(T) + 0.3 \cdot \text{ADD}(T) + 0.3 \cdot |\text{clauses}|$$

$$\text{ADD}(T) = \frac{1}{|E|} \sum_{(i,j) \in E} |i - j|$$

Signal: lower syntactic complexity (C_syn) may indicate cognitive load from depression.

Linguistic features capture psychological indicators: self-focus, reduced syntactic complexity, and medical concerns.

Deep Learning Models

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Background architectures — Transformer (MuRIL) is the primary model

Note: RNN/LSTM are background theory. **The MuRIL Transformer is the actual model used in this project.**

RNN / LSTM

Background architecture

$$h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t)$$

LSTM: Forget / Input / Output gates solve the vanishing gradient problem

$$h_t = o_t \odot \tanh(C_t)$$

Attention Mechanism

Weighting context

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Scaling by $\sqrt{d_k}$ prevents softmax saturation at large dimensions.

Transformer (MuRIL)

Primary model — used in system

$$X' = \text{LayerNorm}(X + \text{MultiHead}(X, X, X))$$

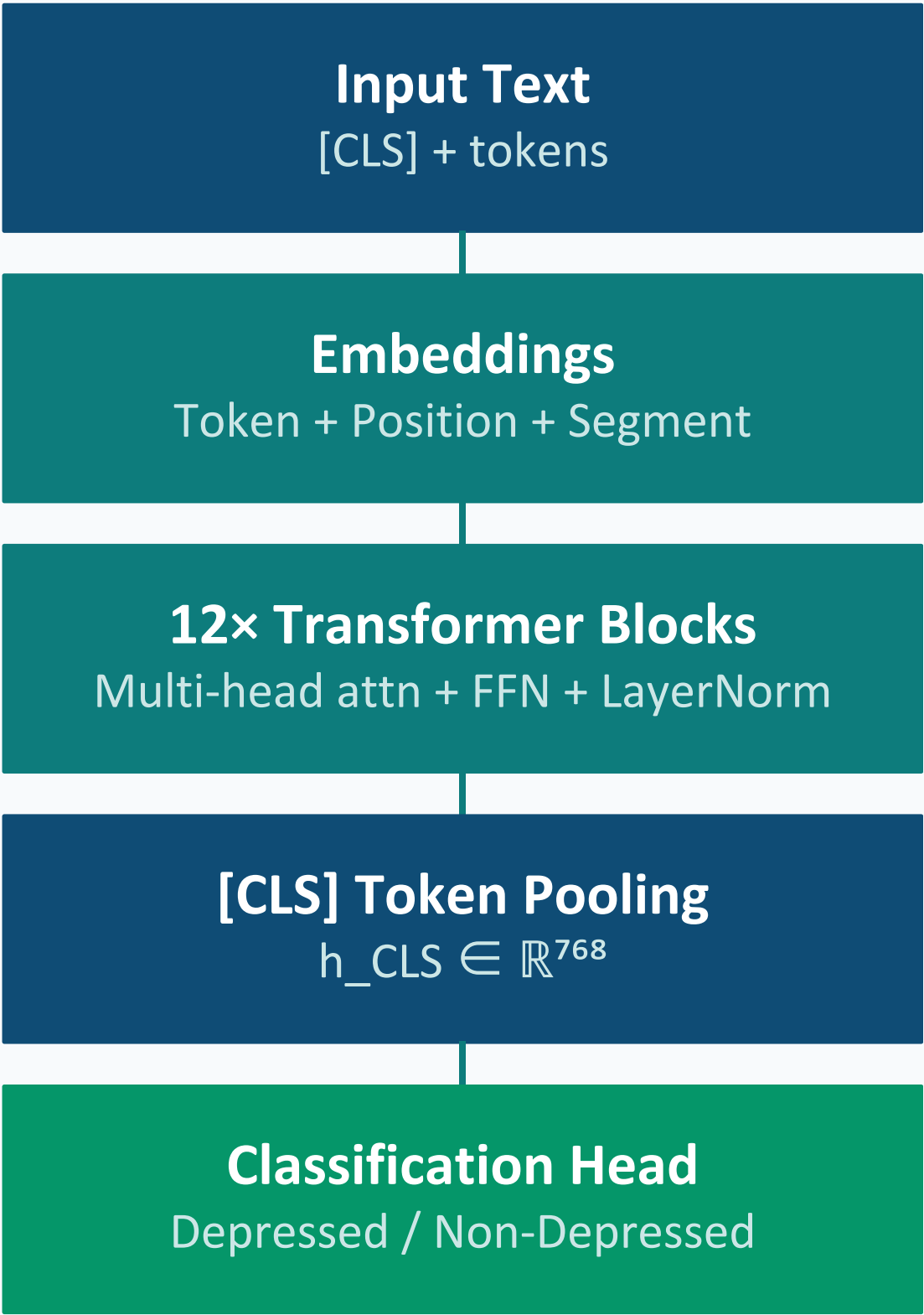
$$X'' = \text{LayerNorm}(X' + \text{FFN}(X'))$$

Positional encoding adds order info.

Positional Encoding:

$$PE_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{2i/d}}\right)$$

$$PE_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{2i/d}}\right)$$



Architecture Configuration	
Vocab size	200,000 tokens (incl. Dravidian scripts)
Layers	12 encoder transformer blocks
Hidden size	768 dimensions
Attention heads	12 heads (dk = 64 per head)
FFN	768 → 3072 → 768 (GELU activation)
Pre-training	17 Indian languages (MLM + NSP)

Training Details

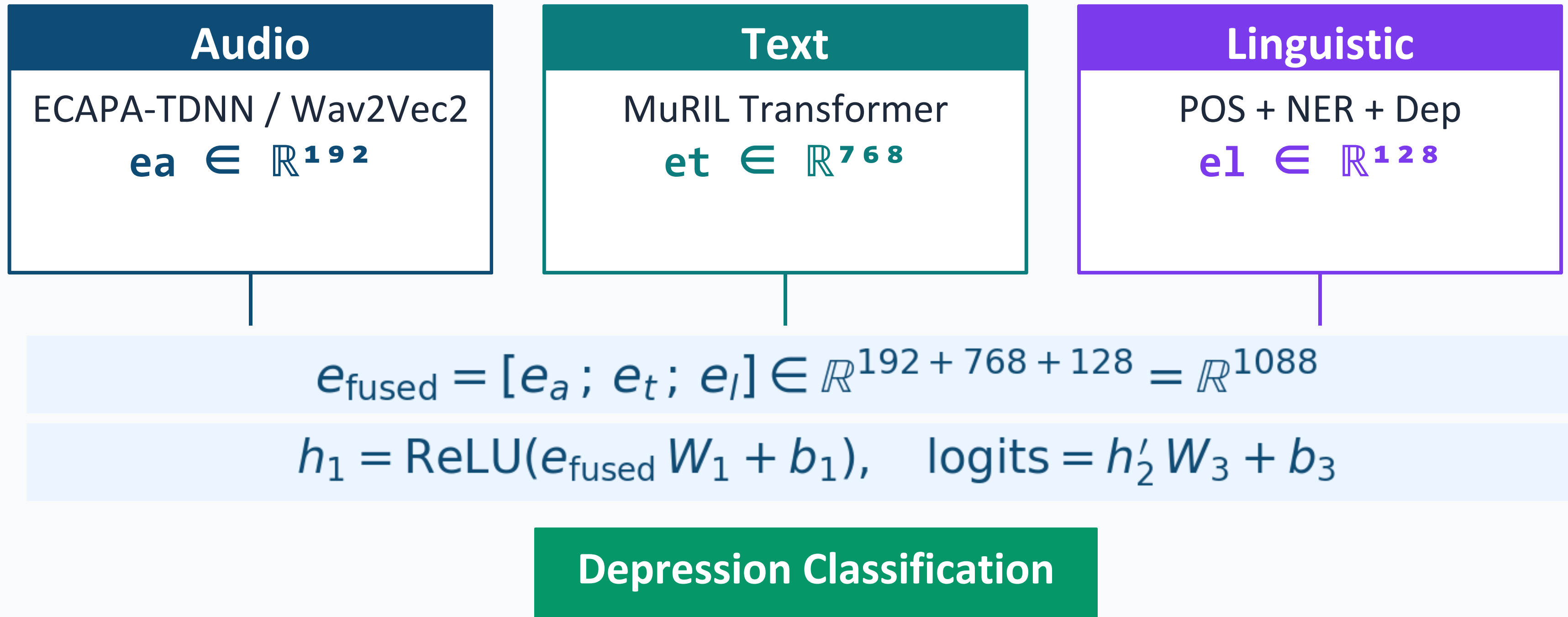
Optimizer: AdamW | lr = 2e-5 | weight_decay = 0.01 | Dropout = 0.1

$$\theta_t = \theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}$$
$$\mathcal{L}_{\text{smooth}} = (1 - \alpha) \mathcal{L} + \frac{\alpha}{K}, \quad \alpha = 0.1$$

Multimodal Fusion Architecture

09

Combining audio, text, and linguistic features



Linguistic feature importance
weighting:

$$w_i = \text{softmax}(\theta_i), \quad e_l^w = e_l \odot w$$

Fusion Strategies

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Three approaches — Gated Fusion used in the final model

Final model uses Gated Fusion — adaptive per-modality weighting learned from data.

Concatenation

Baseline fusion

$$e_{\text{fused}} = [e_a; e_t; e_l] \in \mathbb{R}^{192 + 768 + 128} = \mathbb{R}^{1088}$$

Simple, no extra params.
Equal weight to all modalities.
Macro-F1 = 0.86

Attention-Based

Cross-modal weighting

$$\alpha = \text{softmax}\left(\frac{QK^T}{\sqrt{d_h}}\right) \in \mathbb{R}^{3 \times 3}$$

α_{12} : audio $\rightarrow$ text
 α_{23} : text $\rightarrow$ linguistic
Macro-F1 = 0.87

Gated (Final Model)

Adaptive modality weighting

$$g_m = \sigma(W_m^g e_m + b_m^g)$$

$$e_{\text{fused}} = g_a \odot e'_a + g_t \odot e'_t + g_l \odot e'_l$$

Gates suppress irrelevant modalities dynamically.
Macro-F1 = 0.89

Evaluation Metrics

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Formulas used to measure depression detection performance

Precision

$$P_c = \frac{TP_c}{TP_c + FP_c}$$

Fraction of positive predictions that are correct.

Recall

$$R_c = \frac{TP_c}{TP_c + FN_c}$$

Fraction of actual positives correctly identified.

F1 per Class

$$F1_c = \frac{2 P_c R_c}{P_c + R_c}$$

Harmonic mean of precision and recall per class.

Macro-F1 (primary)

$$F1_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C F1_c$$

Unweighted mean of per-class F1. Handles class imbalance.

Confusion Matrix

	Non-Dep	Dep
Non-Dep	450	50
Dep	60	440

Statistical Validation

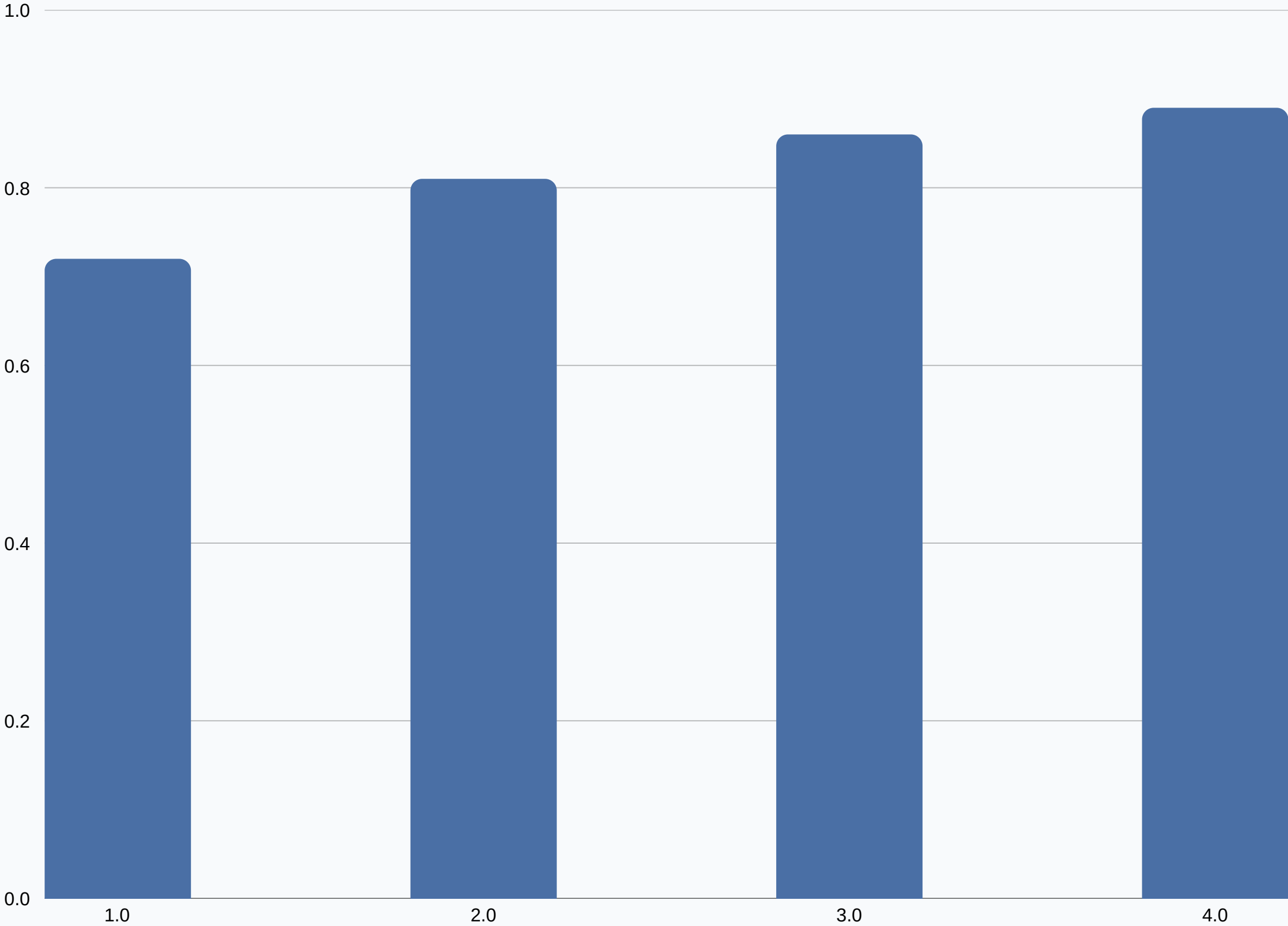
$$CV = \frac{1}{K} \sum_{k=1}^K F1_{\text{macro}}^{(k)}, \quad K = 5$$

$$\chi^2 = \frac{(b - c)^2}{b + c} \quad [\text{McNemar's test}]$$

$$CI_{95\%} = \bar{x} \pm 1.96 \frac{s}{\sqrt{n}}$$

Results & Model Comparison

Progressive improvement with multimodal fusion



Model	F1	Accuracy
TF-IDF + SVM	0.72	0.74
MuRIL Text	0.81	0.83
Audio + Text	0.86	0.87
Enhanced MM	0.89	0.90

+23.6%

improvement over baseline

Ablation: Audio 40% | Text 35% | Linguistic 25%

$$F1_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C F1_c$$

Where and why the model struggles

Code-Mixing Errors

~45%

Tamil-English mixed sentences break tokenization and morphological segmentation.

"vali pain feel paNREn" (Tamil + English mixed sentence)

Dialect Variation

~32%

Regional Tamil (Sri Lankan, Iyengar) and Kerala Malayalam dialects diverge from training data.

Colloquial forms not seen in training corpus – models fail silently.

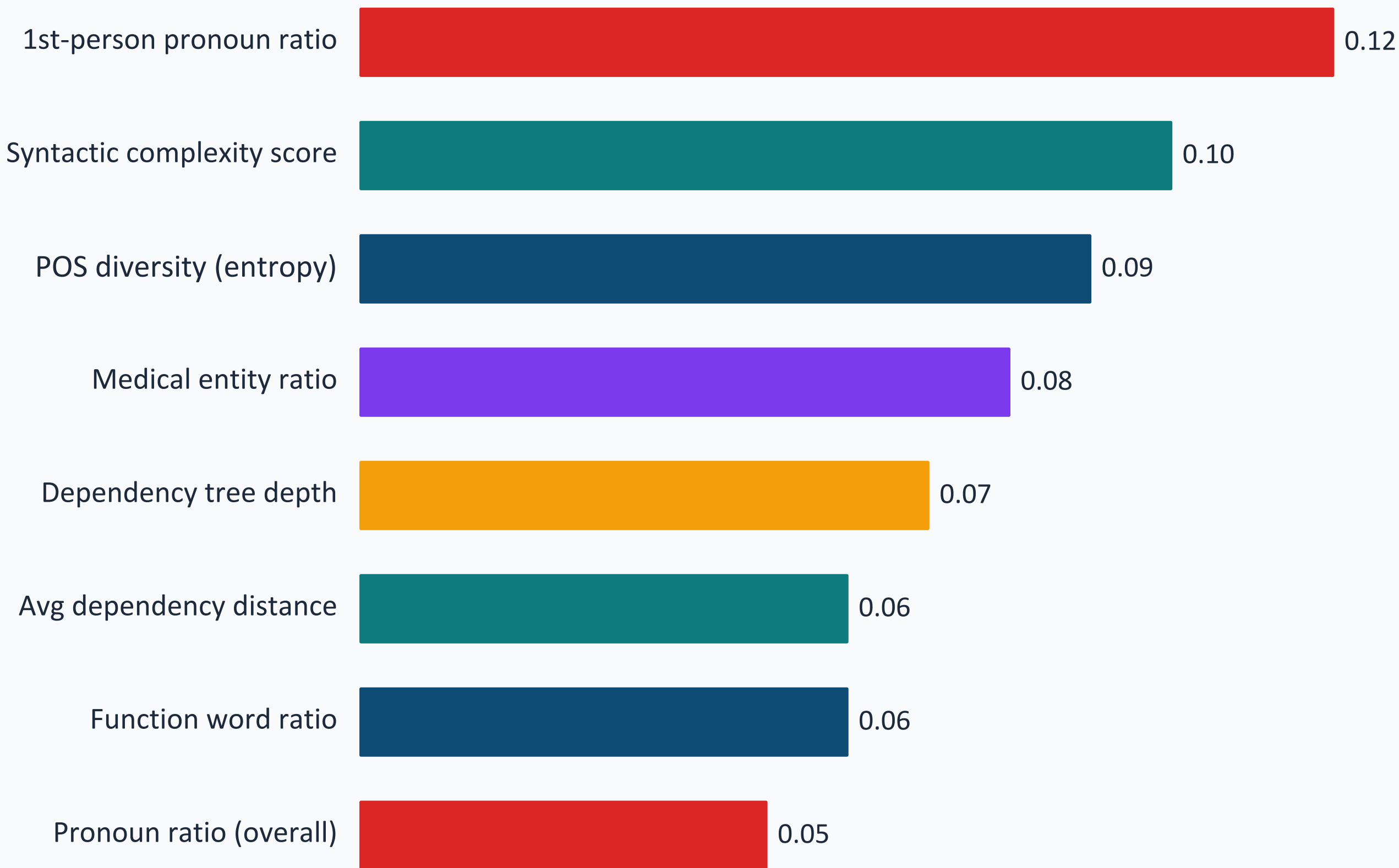
Ambiguous Cases

~23%

Borderline depression expressed through metaphors, indirect speech, or cultural idioms.

"I feel like the monsoon never ends" – culturally ambiguous metaphor

Linguistic Feature Insights



KEY FORMULAS

$$R_{1st} = \frac{|\{w : w \in \text{pronouns}\}|}{|W|}$$

$$C_{syn} = 0.4 \cdot \text{depth}(T) + 0.3 \cdot \text{ADD}(T) + 0.3 \cdot |\text{clauses}|$$

$$R_{med} = \frac{|\text{Medical Entities}|}{|\text{All Entities}|}$$

$$\text{ADD}(T) = \frac{1}{|E|} \sum_{(i,j) \in E} |i - j|$$

Unicode NFC

$$\text{NFC}(s) = \text{compose}(\text{decompose}(s))$$

Morfessor MDL

$$\text{Cost}(M, D) = L(M) + L(D|M)$$

TF-IDF

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \text{IDF}(t, D)$$

SVM Margin

$$\min \frac{1}{2} \|w\|^2 + C \sum \xi_i$$

Transformer Attn

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

CLS Classification

$$\text{logits} = \text{Dropout}(h_{\text{CLS}}) W_c + b_c$$

Concat Fusion

$$e_{\text{fused}} = [e_a; e_t; e_l] \in \mathbb{R}^{1088}$$

Gated Fusion

$$g_m = \sigma(W_m e_m); \quad e_f = g_a \odot e'_a + g_t \odot e'_t + g_l \odot e'_l$$

Cross-Entropy

$$\mathcal{L}_{\text{CE}} = - \sum_i \sum_c y_{ic} \log \hat{y}_{ic}$$

Label Smoothing

$$\mathcal{L}_{\text{smooth}} = (1 - \alpha) \mathcal{L} + \alpha/K$$

Macro-F1

$$F1_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C F1_c$$

AdamW Optimizer

First moment (momentum):

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

Second moment (RMSprop):

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

Parameter update:

$$\theta_t = \theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}$$

$\beta_1=0.9$ | $\beta_2=0.999$ | $\epsilon=1e-8$ | $lr=2e-5$ |
weight_decay=0.01

Regularization Techniques

Dropout (prevents co-adaptation):

$$h_{\text{drop}} = \frac{1}{1-p} (m \odot h), \quad m \sim \text{Bernoulli}(1-p)$$

L2 Regularization:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \sum_i w_i^2, \quad \lambda = 2 \times 10^{-5}$$

Softmax probability output:

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

Additional techniques: early stopping (patience=5),
gradient clipping ($\tau=1.0$), label smoothing ($\alpha=0.1$).

$$\mathcal{L}_{\text{smooth}} = (1 - \alpha) \mathcal{L} + \frac{\alpha}{K}, \quad \alpha = 0.1$$

Data Challenges

PROB Limited labeled data

SOL Transfer learning from MuRIL

PROB Class imbalance

SOL SMOTE oversampling

PROB Code-mixed text

SOL Multilingual MuRIL + language ID

Model Challenges

PROB Overfitting

SOL Dropout 0.3, L2 reg, early stopping

PROB Compute cost

SOL Mixed precision FP16 training

PROB Interpretability

SOL Attention visualization + LIME

Linguistic Challenges

PROB Agglutinative morphology

SOL Morfessor + WordPiece subword

PROB SOV word order

SOL UD dependency parsing

PROB Dialectal variation

SOL Diverse data + normalization

Larger Pre-trained Models

- IndicBERT-v2, mT5, XLM-RoBERTa
- Cross-lingual transfer: Telugu, Kannada
- Zero-shot learning for unseen languages

Real-time Deployment

- Real-time monitoring for telehealth apps
- EHR integration, clinical decision support
- Streaming inference + early warning system

Multimodal Transformers

- Tensor fusion networks
- Cross-modal attention (audio ↔ text)
- Prosodic features: pitch, speech rate, pauses

Discourse & Context Features

- Conversation history context modeling
- Coherence metrics + topic modeling
- Speaker demographic adaptation

01

Text Preprocessing

Unicode NFC normalization + Morfessor MDL morphological segmentation for agglutinative Dravidian text.

02

Word Representations

TF-IDF baseline (F1: 0.72) improved by MuRIL transformer — self-attention across 17 Indian languages (F1: 0.81).

03

Traditional NLP

CRF-based POS tagging, BIO-NER, Arc-standard dependency parsing — yielding 128-dimensional linguistic features.

04

Multimodal Gated Fusion

Audio + Text + Linguistic gated fusion (d=1088) achieves Macro-F1 = 0.89, Accuracy = 0.90 — 23.6% over baseline.

THANK YOU

Questions & Discussion

Multimodal Depression Detection in Dravidian Languages
Team A15 | 22AIE315 NLP | Amrita Vishwa Vidyapeetham

Keerthivasan S V | Krish S | Sriranjana | Thrishala

github.com/S6-Project-Workspace/depression-detection-from-speech

Final Macro-F1: 0.89 | Accuracy: 0.90

IndicWhisper (ASR)

- 769 million parameters
- 10,000 hours of Indian language data
- Transformer-based encoder-decoder architecture with 24 layers.
- Input : Audio files
- Audio → Spectrograms (time–frequency representation) → Remove noise
- **Transcription :**
 - Encoder : create a representation of the speech
 - Decoder generates the corresponding text in the native script

Morfessor MDL (Minimum Description Length)

Segmentation

- Automatically split agglutinative words (e.g., Tamil/Malayalam) into **morphemes** to **reduce vocabulary size** and improve NLP modeling.
- $L(M,D)=L(M)+L(D|M)$ where M = morphological model, D = dataset of words, $L(M)$ = model complexity, and $L(D|M)$ = cost of encoding all words
- algorithm tests possible word splits and selects the segmentation that minimizes

ECAPA - TDNN - 14.7 million parameters

- Emphasized Channel Attention, Propagation and Aggregation – Time Delay Neural Network
- designed for speech analysis
- Produces a 192-dimensional audio feature vector
- TDNN → Capture temp. dependencies
- ECAPA-TDNN improves TDNN using Res2Net modules.
- Input channels → split into groups → processed
- Squeeze-Excitation modules apply channel attention

Wav2Vec2

- **95 million parameters**
- learns features directly from the raw audio waveform
- Speech → CNN feature encoder (7 layers) → captures low level features
- 12 layer transformer network → long range features
- Results are then aggregated

Attention Matrices:

$$W_Q \in \mathbb{R}^{(768 \times 64)}$$

$$W_K \in \mathbb{R}^{(768 \times 64)}$$

$$W_V \in \mathbb{R}^{(768 \times 64)} \quad Q, K, V \in \mathbb{R}^{(n \times 64)}$$